Ensemble Fusion for Inertial-Sensor-Based Field Hockey Activity Recognition

Norazman Shahar
Department of Electrical, Electronic and Systems Engineering, Faculty of Engineering and Built Environment, Universiti Kebangsaan Malaysia (UKM)
Selangor, Malaysia
norazmanshahar@ukm.edu.my

Norhana Arsad
Department of Electrical, Electronic and Systems Engineering, Faculty of Engineering and Built Environment, Universiti Kebangsaan Malaysia (UKM)
Selangor, Malaysia
noa@ukm.edu.my

Ahmad Kamal Ariffin Mohd Ihsan
Department of Mechanical and Manufacturing Engineering, Faculty of Engineering and Built Environment, Universiti Kebangsaan Malaysia (UKM)
Selangor, Malaysia
kamal3@ukm.edu.my

Mohd Faisal Ibrahim
Department of Electrical, Electronic and Systems Engineering, Faculty of Engineering and Built Environment Universiti Kebangsaan Malaysia (UKM)
Selangor, Malaysia
faisal.ibrahim@ukm.edu.my

Aqilah Baseri Huddin
Department of Electrical, Electronic and Systems Engineering, Faculty of Engineering and Built Environment, Universiti Kebangsaan Malaysia (UKM)
Selangor, Malaysia
aqilah@ukm.edu.my

Muhammad Amir As'ari
Department of Biomedical Engineering & Health Sciences, Faculty of Electrical Engineering, Universiti Teknologi Malaysia (UTM)
Johor, Malaysia
amir-asari@utm.my

2025 International Conference on Electrical Engineering and Informatics (ICEEI) | 979-8-3315-9271-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICEEI68459.2025.11330997

***Abstract*— Accurate recognition of complex sports movements remains a challenge in human activity recognition (HAR), particularly in field hockey, where overlapping motion patterns complicate classification. This study evaluates the effectiveness of feature selection and ensemble strategies using multi-sensor inertial data to improve recognition robustness and efficiency. Five feature configurations were developed through hybrid pipelines integrating Minimum Redundancy Maximum Relevance (MRMR) and Regularized Neighborhood Component Analysis (RNCA) and evaluated using polynomial support vector machines (SVM) under a leave-one-subject-out (LOSO) cross-validation framework. The findings revealed that all configurations achieved comparable performance, with accuracies between 85.0% and 85.9% and mean F1-scores ranging from 0.736 to 0.757. The most compact configuration (RNCA2) reduced dimensionality by over 80% while maintaining equivalent accuracy. Statistical analyses confirmed that differences among models were not significant. Additionally, an ensemble based on majority voting between the Original and RNCA configurations further improved stability, yielding an accuracy of 0.860 ± 0.146. These results demonstrate that efficient feature reduction can preserve recognition accuracy, supporting the development of lightweight, subject-independent HAR systems for sports analytics.**

Keywords—Accelerometer, Activity recognition, Gyroscope, leave-subject-one-out, support vector machine

I. INTRODUCTION

HAR utilizing wearable sensors has become an essential technology across various domains, including sports analytics [1, 2], healthcare monitoring [3, 4], and rehabilitation [5]. This approach facilitates the collection and analysis of motion data through inertial measurement units (IMUs), which are devices that generally integrate accelerometers and gyroscopes to capture rich kinematic data [6, 7]. These sensors collaboratively track and measure the movement patterns of an individual, enabling machine learning models to classify these complex movements in real time [8]. In the field of sports, HAR offers valuable insights that enhance performance evaluation [9, 10], technique optimization [11], and injury prevention [12]. By capturing detailed biomechanical signatures, coaches can assess specific actions and movements, identifying areas for improvement that can enhance athletic performance. For example, data gathered during training can illuminate how an athlete executes techniques, aiding in skill refinement.

Nevertheless, recognizing complex sports activities such as passing, dribbling, or tackling in field hockey poses distinct challenges. These activities are often dynamic, non-cyclic, and may overlap, complicating accurate identification compared to simpler daily activities like walking or sitting [13]. The rapid variations in movement and the necessity for precise differentiation between similar actions require advanced algorithms and robust datasets to ensure effective recognition [14]. Therefore, while HAR holds significant promise in sports, ongoing study is crucial to address these challenges and enhance accuracy in complex and real sport applications.

To address the complexities essential in HAR, prior studies have investigated a range of techniques for extracting and selecting handcrafted features, which aim to enhance model performance while reducing computational overhead [14-16]. Among the prominent techniques are MRMR and RNCA. MRMR is designed to identify features that provide the most relevant information while minimizing redundancy [17], thereby preserving key data for accurate classification. Conversely, NCA is a distance metric learning algorithm that can be used for feature selection by learning a feature weighting that emphasizes neighborhood structures, aiding in the removal of irrelevant features [14].

Despite these encouraging developments, many HAR studies have recently focused on controlled, subject-dependent scenarios [14, 18]. In such studies, both the training and testing datasets originate from the same subjects, which can result in overfitting where models excel only in familiar contexts but fail to generalize to new users [19]. This limited focus gives rise to a challenge known as subject-dependence bias, preventing the models from performing effectively in field hockey activity recognition. Consequently, there is a pressing need for study that prioritizes the generalization of HAR models across diverse populations.

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

Authorized licensed use limited to: Universidad de Zaragoza. Downloaded on November 13,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

The exploration of various feature selection strategies through subject-independent validation, particularly utilizing LOSO cross-validation, remains an under-researched area in the context of field hockey. This validation approach is critical, as it provides a more realistic estimate performance of a model to on unseen individuals [20, 21]. The presence of redundant features can significantly compromise the robustness of classification models, whereas overly aggressive feature reduction may hinder recognition performance, especially for minority or overlapping classes.

This study aims to investigate whether compact feature subsets, derived from a hybrid selection pipeline integrating MRMR with RNCA, can maintain accuracy levels comparable to those achieved with the original feature configuration, while also substantially reducing dimensionality and computational costs for subject-independent HAR. These configurations were rigorously tested with polynomial Support Vector Machines (SVM) classifiers to assess their effectiveness. Additionally, ensemble majority-voting models were incorporated to determine whether combining both feature-rich and more compact configurations could enhance recognition performance and improve stability across different subjects. The objective is not only to advance classification accuracy but also to ensure the resulting models are efficient and practical for real-sport applications in field hockey.

II. MEHTODOLOGY

A. Data Preparation and Feature Configurations

The data preparation and feature extraction techniques employed in this study were derived from the authors' previous work and related to study on field hockey activity recognition. The dataset comprised signals collected from four body-mounted inertial sensors, each equipped with a tri-axial accelerometer and gyroscope. From these raw signals, a total of 432 handcrafted features were extracted per sample, which included 14 statistical time-domain descriptors and 4 frequency-domain descriptors. These features effectively encapsulate both the temporal dynamics and spectral characteristics of player motion, and their robustness in sports-related HAR has been validated in prior studies. To assess the impact of feature selection and dimensionality reduction, five feature configurations, as previously stated in earlier studies, were utilized:

1. Original feature configuration – 432 features derived from all sensor signals, serving as the baseline configuration [14].
2. MRMR1 – 216 features selected through the MRMR approach [14].
3. RNCA1 – 215 features selected via RNCA to improve class separability [14].
4. MRMR2 – 222 features selected through MRMR followed by correlation-based filtering to eliminate highly correlated attributes [22].
5. RNCA2 – 83 features generated through a hybrid multi-stage selection method that combines MRMR, correlation filtering, and RNCA, resulting in a compact yet discriminative subset [22].

These feature configurations established a consistent experimental foundation to evaluate how control over feature dimensionality and redundancy influences model accuracy, and computational efficiency in subject-independent field hockey activity recognition.

B. LOSO Cross-Validation Protocol

In each iteration of the LOSO process, data from one subject was designated as the test set, while data from the remaining subjects was combined to form the training set. This procedure was repeated for each subject, with the average performance across all iterations reflecting the overall effectiveness of the model. The LOSO framework provides a robust approach for assessing model generalization to unseen individuals, which is particularly crucial in sports activity recognition tasks due to the significant impact of inter-subject variability on model accuracy.

Within each fold, training features were standardized using z-score normalization. The mean and standard deviation were calculated from the training data and applied to both the training and test sets. This methodology ensured consistent feature scaling and minimized the risk of information leakage from the test set. Each trained model, including SVM and logistic regression, was evaluated independently within the same LOSO framework. This process produced per-subject accuracies, confusion matrices, and per-class F1-scores for comprehensive analysis.

C. Classifier Design and Training Procedure

The classification was performed using a SVM implemented within a multiclass error-correcting output codes (ECOC) framework. The SVM was selected based on its proven capability to handle high-dimensional feature spaces and its consistent performance in prior research on field hockey activity recognition including previous studies conducted by the author [14]. The choice of SVM also aligns with related HAR literature, where it is widely recognized for its stability and strong generalization performance across diverse sensor-based datasets.

A polynomial kernel SVM was adopted to capture nonlinear relationships between the extracted inertial features and the target activity classes. Specifically, a second-order polynomial kernel with a box constraint value of 0.1 was employed, and all input features were standardized prior to training to ensure scale invariance and numerical stability. The kernel function is defined as $\boldsymbol{k}(\boldsymbol{x}_i\, \boldsymbol{x}_j) = (\boldsymbol{x}_i\, \boldsymbol{x}_j + \mathbf{1})^{\boldsymbol{d}}$ where $d = 2$ denotes the polynomial degree. This configuration allows the SVM to model quadratic decision boundaries, offering a balance between expressiveness and computational efficiency.

The ECOC framework decomposed the six-class field hockey recognition problem into multiple binary subproblems, where each binary learner was trained independently, and their aggregated decisions were used to derive the final predicted class. During each LOSO iteration, the SVM was trained using data from all but one subject, while the remaining subject served as an independent test set. This process was repeated for every subject to assess generalization to unseen individuals, a key criterion in sports-related HAR applications. All features were normalized using z-score parameters computed from the training subset and subsequently applied to both training and test data to prevent information leakage.

It is important to note that this study does not aim to optimize or compare classification algorithms. Instead, the SVM served as a consistent and reliable baseline classifier to

Authorized licensed use limited to: Universidad de Zaragoza. Downloaded on November 13,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

isolate the effects of feature dimensionality, feature selection, and ensemble integration under the LOSO evaluation protocol. The primary objective was to observe the influence of different feature configurations and the effectiveness of ensemble strategies in improving model stability and generalization performance across subjects.

D. Performance Evaluation and Statistical Analysis

The performance of each feature configuration was rigorously evaluated utilizing quantitative metrics that assessed both overall accuracy and class-level discrimination capabilities. In the context of each LOSO fold, the trained SVM generated predictions for the held-out subject, which were subsequently compared with the corresponding ground-truth activity labels. To comprehensively characterize recognition performance, we computed per-subject accuracy, confusion matrices, and class-wise F1-scores. Accuracy was defined as the proportion of correctly classified instances relative to the total number of test samples, thereby providing a robust measure of the reliability of the model. The F1-score was employed to evaluate the balance between precision and recall for each class, particularly for activities characterized by unequal sample distributions. This score was derived as the harmonic mean of precision and recall, with precision and recall defined for each specific class. The macro-averaged F1-score across all classes and subjects served as a consolidated indicator of balanced classification performance.

To ascertain the statistical significance of the differences among the five feature configurations, a Friedman test was performed utilizing fold-wise accuracies obtained from each model. This non-parametric procedure is particularly suited for repeated-measures designs in which multiple classifiers are assessed on identical data partitions. The Friedman test ranked each model based on its per-fold accuracy, with the mean rank serving as an aggregate measure of comparative performance. In instances where the Friedman test resulted in a significant outcome ($p < 0.05$), a Nemenyi post-hoc test was conducted to identify which model pairs exhibited significant differences. The analysis revealed no statistically significant differences among the models ($p > 0.05$), thereby suggesting that all evaluated configurations displayed comparable performance within the LOSO framework.

Furthermore, to evaluate the magnitude of pairwise performance differences, Cohen's d effect sizes were calculated between all model pairs. This measure serves to quantify the practical significance of mean accuracy differences, extending beyond conventional hypothesis testing. The results indicated that all effect sizes were negligible ($d < 0.2$), thereby confirming that the observed variations in accuracy were minimal and lacked practical significance. Collectively, the Friedman test and effect size analysis substantiated that the examined feature selection strategies achieved statistically equivalent performance, reinforcing the robustness of the classification framework across diverse feature dimensionalities.

E. Ensemble Evaluation (Majority Voting)

In addition to individual feature-based models, this study further examined whether combining complementary feature representations could enhance classification robustness. To this end, several ensemble configurations were constructed using a majority voting strategy that integrated the predictions of independently trained Support Vector Machine (SVM) models. The rationale behind this approach was that different feature configurations capture distinct aspects of motion dynamics some emphasizing global signal characteristics, while others capture compact, discriminative subsets and their combination may yield improved stability and generalization across subjects.

Three ensemble variants were investigated: the combination of original and RNCA2, original and RNCA, and RNCA with RNCA2. Each ensemble was evaluated under the same Leave-One-Subject-Out (LOSO) framework to ensure subject-independent validation. For each fold, both base SVM classifiers were trained separately using their respective feature sets and identical hyperparameter settings. Their individual predictions on the test subject were then aggregated through a simple majority voting rule, whereby the most frequently predicted label among the constituent models was selected as the final output. This procedure maintained the interpretability of the ensemble and avoided introducing additional complexity or tuneable parameters.

The ensemble models were evaluated using the same performance metrics as the individual configurations, including per-subject accuracy and class-wise F1-scores. Across all ensemble combinations, results demonstrated comparable or slightly improved accuracies relative to single-model baselines, indicating that ensemble voting contributed to enhanced model stability. Among the tested ensembles, the combination of the Original and RNCA configurations achieved the highest mean accuracy and consistent F1 performance across most activity classes. This improvement reflects the complementary strengths of the two feature sets where the original configuration preserves global variability, while the RNCA subset enhances class separability through refined feature weighting.

Overall, the ensemble evaluation confirmed that majority voting offers a simple yet effective means of improving robustness in subject-independent human activity recognition without increasing computational burden. By combining feature-rich and compact representations, the ensemble approach provided a balanced trade-off between accuracy and efficiency, reinforcing its suitability for real-time sports analytics applications.

III. Result and Discussion

A. Overall Performance and Statistical Comparison

The performance outcomes of the five feature configurations are comprehensively summarized in terms of mean accuracy, standard deviation, and mean F1-score, as outlined in Table 1. Across all configurations, the classification accuracy exhibited a range from 85.0% to 85.9%, accompanied by standard deviations that varied between 12.8% and 15.9%. The mean F1-scores were observed to range from 0.736 to 0.757, indicating a consistent level of predictive capability across the evaluated configurations. Collectively, these results demonstrate that all configurations achieved comparably effective predictive performance, with only marginal differences identified among the various feature sets. The original configuration, which employed the entirety of the 432-features, attained an accuracy of 85.4% ± 15.9% and a mean F1-score of 0.755. While this finding establishes a robust baseline for performance, the relatively high standard deviation underscores significant variability in model performance across different subjects. The inconsistency may be attributed to redundancy within the comprehensive feature set, where the

Authorized licensed use limited to: Universidad de Zaragoza. Downloaded on November 13,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

presence of highly correlated or less informative features could introduce peripheral noise, thereby impairing generalization for specific participants. Such considerations justify further investigation to optimize feature selection and enhance the overall predictive capability of the model.

TABLE I. OVERALL PERFORMANCE OF EVALUATED CONFIGURATIONS

Feature configuration	No. of Features	Accuracy (Mean ± SD)	Mean F1-Score
Original features	432	85.4% ± 15.9%	0.755
MRMR1	216	85.9% ± 13.4%	0.747
RNCA1	215	85.9% ± 13.4%	0.752
MRMR2	222	85.3% ± 12.8%	0.736
RNCA2	83	85.0% ± 13.7%	0.757

The MRMR1 configuration, which effectively reduced the feature space to 216 distinct features, demonstrated the highest observed accuracy of 85.9% ± 13.4%, in conjunction with a mean F1-score of 0.747. This finding underscores the efficacy of eliminating redundant features while preserving those most pertinent to the activity labels, implying that such a reduction did not detrimentally impact overall classification performance. On the contrary, the decreased dimensionality expected enhanced the generalization of the model capabilities by addressing the curse of dimensionality, a phenomenon that can lead to overfitting in high-dimensional spaces. Furthermore, the lower standard deviation relative to the original feature set indicates a notable increase in stability across different subjects, a characteristic routinely associated with effective feature selection methods.

Subsequently, the RNCA1 was implemented following the MRMR1 approach, leading to a refined feature set of 215 features. Remarkably, this configuration preserved the same mean accuracy of 85.9% ± 13.4% while yielding a slightly elevated mean F1-score of 0.752. This incremental improvement suggests that RNCA1 was able to successfully reweight the remaining features, thereby enhancing class separability, particularly regarding the delicate balance between precision and recall across various classes. However, the similarity in overall performance between MRMR1 and RNCA1 may indicate that MRMR had already captured most of the relevant signal, with RNCA1 providing only marginal enhancements to the classification capability.

Conversely, the MRMR2 configuration, which included 222-features, achieved an accuracy of 85.3% ± 12.8% and recorded the lowest mean F1-score at 0.736. While the reduced variance, as evidenced by the lowest standard deviation, suggests a greater level of consistency across subjects, the decline in the mean F1-score indicates a possible loss of crucial discriminative features, particularly those corresponding to minority classes. This may be attributed to the correlation filtering process, which appears to have eliminated features that, despite their correlation, still contributed uniquely to class separability within a multiclass framework. This observed trade-off between stability and discriminative richness may elucidate the slightly diminished F1 performance observed in this configuration.

Finally, the RNCA2 configuration, distinguished by its extremely compact set of only 83 features, attained an accuracy of 85.0% ± 13.7% and, notably, the highest mean F1-score of 0.757. This illustrates that a discriminatingly selected, minimal feature set can not only meet but may also slightly surpass the performance of the original feature set in terms of balanced classification. The relatively high F1-score highlights improved performance across individual classes, suggesting that the RNCA2 subset effectively captured the most salient discriminative features while judiciously discarding redundant attributes. The observed minor decrease in accuracy when compared to MRMR2 may stem from the aggressive feature reduction that led to the exclusion of some marginally useful features. Nonetheless, the substantial efficiency gains exceeding an 80% reduction in dimensionality underscore the effectiveness of this compact feature representation.

From this analysis, two broader observations can be derived. First, the comparable performance across all configurations implies that HAR models exhibit a robust tolerance to dimensionality reduction, given that selection methodologies prioritize the preservation of the most informative features. Second, the general trend of reduced variability across subjects associated with feature reduction reflects more stable generalization capabilities. The slight trade-off between accuracy and F1-score across various methods highlights the distinct focuses of the selection algorithms. MRMR prioritizes overall discriminative relevance, while RNCA enhances class separability, which may particularly benefit minority or challenging-to-separate classes.

B. Analysis Per-Class Mean F1-Scores

While the overall accuracies across the five feature configurations appeared to be similar, the per-class F1-scores revealed significant differences in classification difficulty and model specialization as illustrated in Figure 1(refer to Table 2). Activities such as dribbling and tackling consistently achieved the highest F1-scores, surpassing 0.84 in all configurations. These findings suggest that both activities exhibit distinct kinematic patterns that are reliably distinguishable from other field hockey movements. Dribbling is characterized by continuous and repetitive stick-handling motions that produce unique signatures in accelerometer and gyroscope data, whereas tackling is defined by forceful, abrupt movements with specific temporal and spatial dynamics. Furthermore, the relatively high representation of these activities in the dataset may have enhanced classifier stability for them.

In contrast, the activities of drag flicking and receiving were found to be significantly more challenging, as evidenced by their F1-scores, which ranged from 0.638 to 0.698 for drag flicks and from 0.483 to 0.591 for receiving. Several factors may elucidate this increased difficulty. First, both activities exhibit biomechanical similarities with other actions for instance, the drag flick shares initial body positioning with passing and driving, while receiving mirrors the motion dynamics of dribbling and tackling. Second, the dataset demonstrated a notable class imbalance, characterized by a lower number of samples associated with drag flicks and receiving compared to the more frequently represented actions such as passing and dribbling. This imbalance likely skewed the classifiers towards the majority classes, thereby diminishing recall for the underrepresented activities.

Authorized licensed use limited to: Universidad de Zaragoza. Downloaded on November 13,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

Consequently, the interplay of subtle inter-class differences and unequal class distributions accounts for the comparatively lower F1-scores observed.

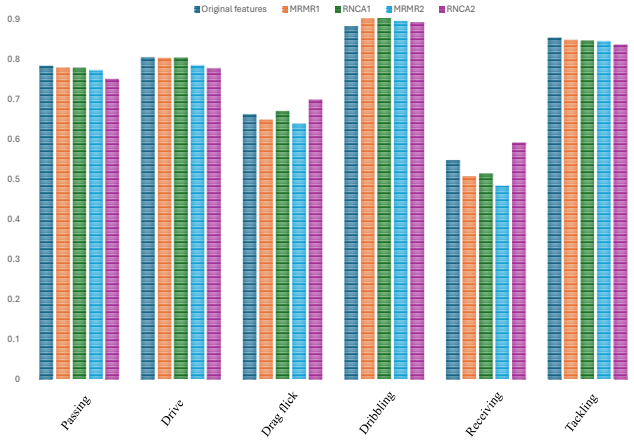


Fig. 1. F1-Scores Per-Class.

The original feature configuration comprising 432-features yielded optimal performance for the actions of passing, driving, and tackling. This outcome underscores the utility of retaining a comprehensive array of features, inclusive of both time-domain and frequency-domain descriptors, which proves advantageous for activities with complex yet well-defined signal profiles. The incorporation of correlated features, while potentially redundant in various contexts, may have preserved critical cues necessary for effective discrimination among these common actions. Remarkably, the RNCA2 configuration, which is comprised of only 83 features, outperformed all other models specifically for the more challenging activities of drag flicking and receiving, despite achieving a dimensionality reduction of over 80%. This finding suggests that the RNCA2 configuration successfully identified the most salient discriminative features while eliminating redundant or noisy variables that could obscure subtle patterns within minority classes. By enhancing the signal-to-noise ratio, the compact RNCA2 feature set improved class separability in contexts where activities exhibited significant overlap. This outcome highlights the potential of aggressive feature reduction strategies for addressing class imbalance and enhancing performance related to minority classes.

Although these configurations did not surpass RNCA2 in terms of recognizing drag flicks and receiving, their consistent performance across all six activities suggests that moderate dimensionality reduction may be advantageous for achieving robust overall performance without compromising interpretability. These findings illustrate that classification difficulty in field hockey activity recognition is strongly dependent on class characteristics. Kinematically distinct activities such as dribbling and tackling demonstrate high reliability in classification, irrespective of feature configuration. Conversely, less frequently occurring and overlapping activities like drag flicking and receiving necessitate selective feature reduction to attain competitive performance levels. These findings underscore the importance of per-class analyses that extend beyond global accuracy metrics and suggest that future research could benefit from employing ensemble approaches or class-dependent weighting strategies that directly address issues of data imbalance.

C. *Statistical Comparison of Model Configurations*

Figure 2 presents the findings of the Friedman test, which was conducted to assess the differences in performance among the various models evaluated. The results indicate that there are no statistically significant differences between the models, as evidenced by a p -value greater than 0.05. This outcome suggests that the per-subject accuracies exhibited by the models were largely comparable across the LOSO folds, indicating a robust level of consistency across the evaluated configurations. In addition, the average ranks derived from the Friedman analysis were found to be nearly identical for all five model configurations. The plotted values exhibit a high degree of overlap, further reinforcing the observation that the performance metrics are closely aligned. This uniformity in average ranking is indicative of the narrow range of mean accuracies previously reported ranging from 85.0% to 85.9%.

These findings collectively demonstrate that the selection of feature extraction and selection strategies had a negligible effect on the overall ranking of the models. Thus, while variations exist in the individual accuracies of each configuration, they do not translate into significant differences in model performance or efficacy as captured by the Friedman's test analysis. This further underscores the importance of contextualizing feature selection techniques within the broader framework of model evaluation, as their influence on global performance metrics appears to be minimal.

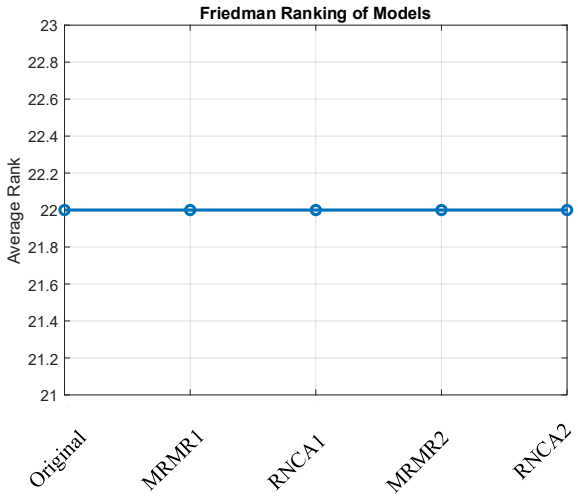


Fig. 2. Friedman ranking of configurations.

Notably, none of the pairwise rank differences surpassed the Nemenyi critical difference, indicating that no single configuration could be regarded as consistently superior at the 95% confidence level. The uniform distribution of ranks suggests that all models, irrespective of their dimensionality, exhibited essentially equivalent performance when evaluated within a subject-independent framework. This finding is particularly noteworthy, as it suggests that aggressive dimensionality reduction can be implemented without incurring significant degradation in predictive accuracy. For instance, RNCA2 achieved a reduction in the feature set of approximately 80%, decreasing from 432 features to 83, while still maintaining performance comparable to that of the original configuration.

Authorized licensed use limited to: Universidad de Zaragoza. Downloaded on November 13,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

The implications of these results are twofold. First, the reinforce the robustness of the underlying classification framework: the polynomial-kernel Support Vector Machine (SVM) appears resilient to moderate alterations in feature set size, provided that critical discriminative features are preserved. Second, the underscore the practical advantages of feature selection, as more compact configurations can yield efficiency gains in terms of computational costs and storage requirements, all while sustaining accuracy on par with the original, high-dimensional set. Consequently, the Friedman ranking analysis supports the conclusion that the evaluated feature selection methodologies of MRMR and RNCA are statistically interchangeable. The selection among these methods may be more influenced by computational considerations or class-specific strengths rather than by overall accuracy.

Additionally, the pairwise effect size analysis utilizing Cohen's d provided further corroboration of the equivalence among the evaluated feature configurations. As illustrated in Figure 3, the heatmap demonstrated a uniformly dark blue pattern, signifying that all pairwise effect sizes approached zero. Quantitatively, the typical values were significantly below the conventional threshold for a "small" effect ($d = 0.20$), with most comparisons yielding $d < 0.10$. This indicates that, although marginal numerical differences in mean accuracy were noted among configurations such as MRMR, RNCA, and RNCA2, the magnitude of these differences was practically negligible.

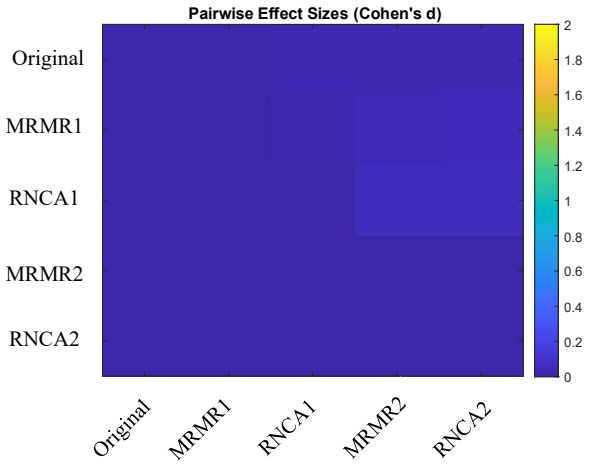


Fig. 3. Pairwise effect sizes (Cohen's d).

The absence of meaningful effect sizes confirms that no configuration offers a consistent or substantive advantage over others. Even in instances where MRMR or RNCA produced slightly higher average accuracies, the variability across subjects rendered these advantages unstable, resulting in trivial effect sizes. These findings are consistent with the non-significant Friedman test, further reinforcing the conclusion that the five feature configurations perform equivalently within the Leave-One-Subject-Out (LOSO) evaluation framework. Methodologically, the negligible effect sizes highlight the robustness of the classification framework, underscoring that dimensionality reduction did not compromise model reliability. This outcome is especially critical for applications in human activity recognition, where computational efficiency and generalization across subjects are paramount. The results suggest that compact feature sets, such as RNCA2, can be selected for their efficiency without forfeiting statistical or practical performance, as their differences relative to larger configurations are indistinguishable in terms of effect size. These results provide compelling evidence that the various feature selection strategies assessed in this study are functionally equivalent regarding overall accuracy when subjected to a rigorous subject-independent protocol. While descriptive trends may indicate that certain models are marginally more suitable for specific classes or contexts, global comparisons confirm that none possesses a statistically significant advantage.

D. Ensemble – Majority Voting

Table 3 summarizes their respective accuracies and per-class F1-scores. Across all ensembles, overall accuracy remained comparable, ranging from 0.854 ± 0.150 to 0.860 ± 0.146, with standard deviations slightly lower than those of individual configurations. This indicates that ensemble voting successfully stabilized subject-wise variability by combining complementary decision boundaries from multiple classifiers.

TABLE II. EVALUATION OF ENSEMBLE MAJORITY-VOTING MODELS ACROSS SIX FIELD HOCKEY ACTIVITIES

	Majority Voting		
	Original configuration and RNCA2	Original configuration and RNCA	RNCA and RNCA2
Accuracy (%)	0.854 ± 0.150	0.860 ± 0.146	0.857 ± 0.135
F1-Scores (passing)	0.778	0.784	0.769
F1-Scores (drive)	0.779	0.808	0.779
F1-Scores (drag flick)	0.692	0.661	0.679
F1-Scores (dribbling)	0.895	0.899	0.905
F1-Scores (receiving)	0.593	0.571	0.592
F1-Scores (tackling)	0.842	0.851	0.839

Among the three ensembles, the combination of the Original model and RNCA achieved the highest mean accuracy of 0.860 ± 0.146, along with superior F1-scores across the majority of activities, particularly in driving (0.808) and tackling (0.851). This suggests that integrating the comprehensive feature set of the Original model with the discriminatively weighted subset from RNCA effectively captured both global patterns and class-specific characteristics.

The RNCA + RNCA2 ensemble, while slightly lower in accuracy at 0.857 ± 0.135, produced the highest F1-score for dribbling at 0.905, confirming that compact, redundancy-minimized features are highly effective for distinctly recognizable activities. Conversely, the Original + RNCA2 ensemble maintained a balanced performance across all classes but did not surpass the other two combinations. This indicates that the further reduction of correlated features may have slightly limited its precision for minority classes.

A per-class analysis revealed that the ensemble methods maintained the same class-dependent trends observed in the

Authorized licensed use limited to: Universidad de Zaragoza. Downloaded on November 13,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.

individual models. Activities such as dribbling and tackling consistently achieved the highest F1-scores ($\geq$ 0.84), reflecting their distinctive kinematic signatures. In contrast, drag flick and receiving were the most challenging, with moderate F1-scores of 0.66–0.69 and 0.57–0.59, respectively, likely due to motion overlap and class imbalance. Nevertheless, the integration of ensembles resulted in a marginal improvement for these difficult classes compared to single-model outcomes, signifying better generalization through consensus-based voting.

Overall, the evaluation of the ensembles demonstrates that majority voting can modestly enhance accuracy and reduce variability without increasing model complexity. The complementary strengths of the Original and RNCA configurations yielded the most stable and balanced recognition performance, underscoring the benefits of combining feature-rich and compact representations for robust, subject-independent human activity recognition in field hockey..

IV. Conclusion

This study evaluated multiple feature selection and ensemble strategies for field hockey activity recognition using inertial sensor data under a subject-independent framework. The results demonstrated that all five feature configurations ranging from the original 432-feature set to the compact 83-feature RNCA2 subset achieved statistically equivalent performance, with accuracies around 85% and stable F1-scores across classes. The non-significant Friedman test and near-zero effect sizes confirmed that dimensionality reduction through MRMR and RNCA does not degrade model performance. Notably, the RNCA2 configuration achieved substantial efficiency gains with minimal accuracy loss, highlighting its potential for resource-constrained or real-time applications. Ensemble models combining the Original and RNCA feature sets further improved performance stability, validating the advantage of fusing complementary feature representations. Overall, these findings underscore that HAR systems can maintain high recognition accuracy even with aggressive feature reduction, enabling efficient, scalable, and generalizable implementations for sports analytics and athlete monitoring.

Acknowledgment

The authors gratefdully acknowledge the financial support from the Ministry of Higher Education (MoHE). This research was funded by MoHE through the Inisiatif Belanjawan 2025: AI Enhancement with grant number WARISAN-2025-023 for their unwavering support which have been instrumental in the completion of this work.

References

[1] T. S. Qureshi, M. H. Shahid, A. A. Farhan, and S. Alamri, "A systematic literature review on human activity recognition using smart devices: advances, challenges, and future directions," *Artificial Intelligence Review,* vol. 58, no. 9, p. 276, 2025.

[2] H. Kaur, V. Rani, and M. Kumar, "Human activity recognition: A comprehensive review," *Expert Systems,* vol. 41, no. 11, p. e13680, 2024.

[3] F. Serpush, M. B. Menhaj, B. Masoumi, and B. Karasfi, "Wearable sensor-based human activity recognition in the smart healthcare system," *Computational intelligence and neuroscience,* vol. 2022, no. 1, p. 1391906, 2022.

[4] E. Fridriksdottir and A. G. Bonomi, "Accelerometer-based human activity recognition for patient monitoring using a deep neural network," *Sensors,* vol. 20, no. 22, p. 6424, 2020.

[5] A. Moawad, M. A. El-Khoreby, S. I. Fawaz, H. H. Issa, M. I. Awad, and A. Abdellatif, "New Framework for Human Activity Recognition for Wearable Gait Rehabilitation Systems," *Applied System Innovation,* vol. 8, no. 2, p. 53, 2025.

[6] S. Subramaniam, A. I. Faisal, and M. J. Deen, "Wearable sensor systems for fall risk assessment: A review," *Frontiers in digital health,* vol. 4, p. 921506, 2022.

[7] R. J. Mobbs *et al.*, "Gait metrics analysis utilizing single-point inertial measurement units: A systematic review," *Mhealth,* vol. 8, p. 9, 2022.

[8] A. Raza, M. R. Al Nasar, E. S. Hanandeh, R. A. Zitar, A. Y. Nasereddin, and L. Abualigah, "A novel methodology for human kinematics motion detection based on smartphones sensor data using artificial intelligence," *Technologies,* vol. 11, no. 2, p. 55, 2023.

[9] A. Alzahrani and A. Ullah, "Advanced biomechanical analytics: Wearable technologies for precision health monitoring in sports performance," *Digital Health,* vol. 10, p. 20552076241256745, 2024.

[10] S. Mekruksavanich, W. Phaphan, N. Hnoohom, and A. Jitpattanakul, "Recognition of sports and daily activities through deep learning and convolutional block attention," *PeerJ Computer Science,* vol. 10, p. e2100, 2024.

[11] X. Suo, W. Tang, L. Mao, and Z. Li, "Digital human and embodied intelligence for sports science: advancements, opportunities and prospects," *The Visual Computer,* vol. 41, no. 4, pp. 2477-2493, 2025.

[12] M. Kovoor, M. Durairaj, M. S. Karyakarte, M. Z. Hussain, M. Ashraf, and L. P. Maguluri, "Sensor-enhanced wearables and automated analytics for injury prevention in sports," *Measurement: Sensors,* vol. 32, p. 101054, 2024.

[13] N. Shahar, N. Ghazali, M. As'Ari, and T. Swee, "Wearable inertial sensor for human activity recognition in field hockey: influence of sensor combination and sensor location," in *Journal of Physics: Conference Series*, 2020, vol. 1529, no. 2: IOP Publishing, p. 022015.

[14] N. Shahar, M. A. As'Ari, T. T. Swee, and N. F. Ghazali, "Optimal Activity Recognition Framework Based on Improvement of Regularized Neighborhood Component Analysis (RNCA)," *IEEE Access,* vol. 12, pp. 113300-113313, 2024.

[15] F. Amjad, M. H. Khan, M. A. Nisar, M. S. Farid, and M. Grzegorzek, "A comparative study of feature selection approaches for human activity recognition using multimodal sensory data," *Sensors,* vol. 21, no. 7, p. 2368, 2021.

[16] D. Thakur and S. Biswas, "Guided regularized random forest feature selection for smartphone based human activity recognition," *Journal of Ambient Intelligence and Humanized Computing,* vol. 14, no. 7, pp. 9767-9779, 2023.

[17] H. Fang, P. Tang, and H. Si, "Feature selections using minimal redundancy maximal relevance algorithm for human activity recognition in smart home environments," *Journal of Healthcare Engineering,* vol. 2020, no. 1, p. 8876782, 2020.

[18] D. Khan *et al.*, "Robust human locomotion and localization activity recognition over multisensory," *Frontiers in Physiology,* vol. 15, p. 1344887, 2024.

[19] S. Pe, G. Nicora, B. M. V. Guerra, S. Sozzi, and E. Parimbelli, "Systematic Comparison of Machine Learning for Activity Recognition in Cross-Subject vs. NonCross-Subject Scenarios: A Preliminary Analysis," in *2024 IEEE 8th Forum on Research and Technologies for Society and Industry Innovation (RTSI)*, 2024: IEEE, pp. 375-379.

[20] D. Gholamiangonabadi, N. Kiselov, and K. Grolinger, "Deep neural networks for human activity recognition with wearable sensors: Leave-one-subject-out cross-validation for model selection," *Ieee Access,* vol. 8, pp. 133982-133994, 2020.

[21] S. U. Rehman, A. Ali, A. M. Khan, and C. Okpala, "Human Activity Recognition: A Comparative Study of Validation Methods and Impact of Feature Extraction in Wearable Sensors," *Algorithms,* vol. 17, no. 12, p. 556, 2024.

[22] N. Shahar, "Inertial Sensor-Based Recognition of Field Hockey Activities Using A Hybrid Feature Selection Framework," Under Review.

Authorized licensed use limited to: Universidad de Zaragoza. Downloaded on November 13,2026 at 12:05:43 UTC from IEEE Xplore. Restrictions apply.